Vectorized Interdisciplinary Overlap between STEM Domains

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Multidisciplinary Motivation

- Students exercise *agency* in course selection beyond their declared major
- Cross-departmental enrollment signals overlap between programs
- NSF identifies convergence research as a top 10 BIG Idea: grand challenges "will not be solved by one discipline alone"
- Interdisciplinary programs can broaden participation for students with marginalized identities

What is MIDFIELD?

- Multiple-Institution Database for Investigating Engineering Longitudinal Development
- Academic records of 98,000+ students from 18 US universities
- Data from 2000 onward to reflect contemporary curricula
- Contains: degree records, course enrollment, grades, demographics

Contribution to the Field

- Shift unit of analysis from *student pathways* to *program relationships*
- Student enrollment as expression of agency and interdisciplinary interest
- Complementary similarity metrics applied to enrollment vectors
- Define three interpretable relationship types:
  - Nested: one program's coursework is largely contained within another
  - Parallel: shared base, divergent specialization
  - Convergent: deep overlap in both general and specialized content
- Extend curricular analytics *across* disciplinary boundaries

Research Question

RQ: What types of curricular relationships have emerged between undergraduate STEM programs as evidenced by student enrollment patterns?

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From Student Records to Program Vectors

1. Identify all courses taken by students completing each CIP code
2. Count course frequencies across all graduates of that major
3. Result: a high-dimensional *enrollment vector* per program
4. Apply similarity metrics to compare vectors pairwise

Why Multiple Metrics?

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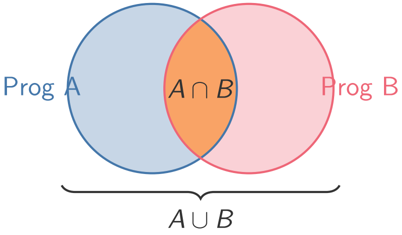
- A single metric cannot distinguish:
  - Broad gen-ed overlap vs. specialized content overlap
  - Subset containment vs. symmetric overlap
  - Expected vs. statistically surprising co-enrollment

Jaccard Index (Curricular Width)

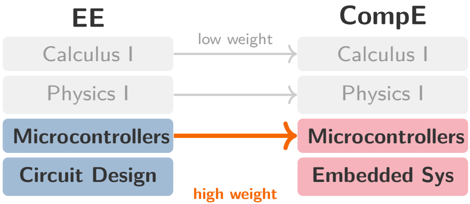


$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

How much total curricular territory do two programs share?

- Treats every course equally
- Diluted by rare electives in large programs

TF-IDF Cosine (Specialized Identity)

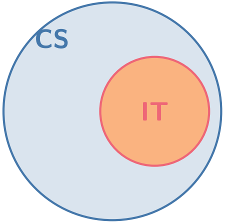


Do two programs share identity-defining coursework?

- Weights use TF-IDF: upweights courses unique to few programs; downweights ubiquitous ones
- Resistant to program size differences
- Isolates specialized content from gen-ed noise

$$\cos(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|\,|\vec{b}|}$$

Overlap Coefficient (Subset Detection)



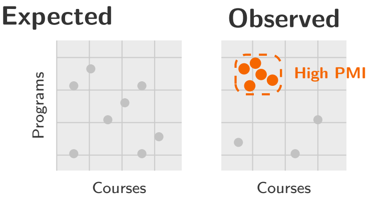
Overlap $\approx$ 80%
Jaccard $\approx$ 8.5%

$$\text{Overlap}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}$$

Is one program's curriculum largely a subset of another's?

- Divides by the *smaller* set
- High Overlap + Low Jaccard = nested

PMI (Signature Bonds)

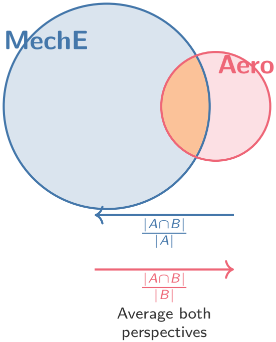


$$\mathrm{PMI}(m, c) = \log \frac{p(m, c)}{p(m)\, p(c)}$$

Do two programs share statistically surprising courses?

- Compares observed co-occurrence to expected
- High PMI = courses that "belong together"

Kulczynski (Mutual Resemblance)



How well would a student from one program fit in the other?

- Averages overlap from *each* perspective
- Robust for large vs. small programs

$$\mathrm{Kulc}(A, B) = \frac{1}{2}\left(\frac{|A \cap B|}{|A|} + \frac{|A \cap B|}{|B|}\right)$$

Relationship Taxonomy

Table: Metric signatures for three curricular relationship types

Type	Jaccard	TF-IDF	Overlap	PMI	Example
Nested	Low	Low	High	Var.	IT $\subset$ CS
Parallel	Med	Low	Med	Low	EE $\parallel$ Civil Eng
Convergent	High	High	High	High	Aero $\approx$ Ind Eng

- The *pattern* across metrics defines the relationship, not any single score
- Relative values within a domain matter more than absolute percentages

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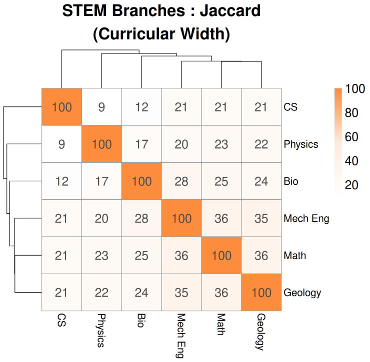
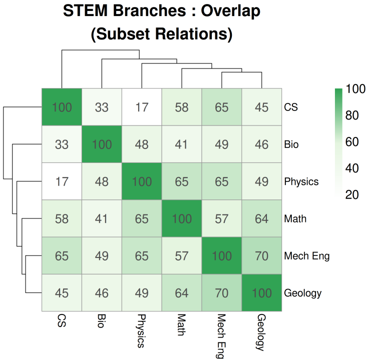
STEM Overview: Mathematics as a Bridge



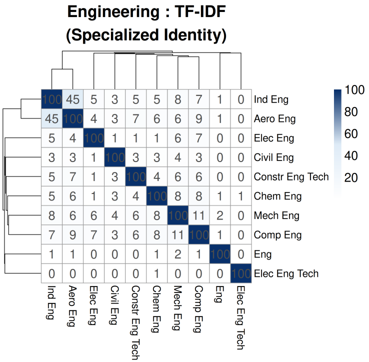
- Math shares coursework with Mech Eng (~36%), Physics (~23%), CS (~21%)
- *Generalist connector* across STEM
- Validates the approach

STEM Overview: Nested Patterns Emerge



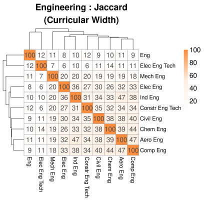
- Physics–Mech Eng: ~65% Overlap vs. ~20% Jaccard
- High Overlap + Low Jaccard = nested relationship
- Physics coursework is largely contained within Mech Eng
- A single metric would miss this asymmetry

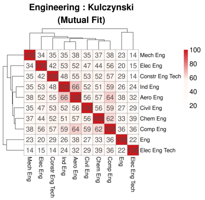
Engineering: The Parallel Pattern



- Median TF-IDF across engineering: ~3%
- Engineering programs share a large common core...
- ...but specialized content is *very* different
- This is the parallel pattern: shared base, divergent identity
- Exception: EE–CompE share ~33% Jaccard + high TF-IDF (merged departments)

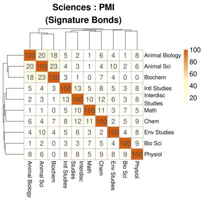
Engineering: Aerospace–Industrial Convergence

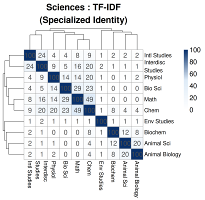




- Flagship result: Aero–Industrial scores highest on *every* metric:
  - TF-IDF ~45%, Jaccard ~47%
  - Kulczynski ~66%, Overlap ~79%
- Textbook convergent relationship
- No one groups these administratively

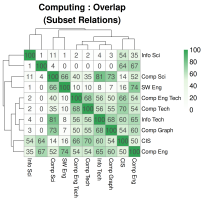
Sciences: Cross-CIP Clustering

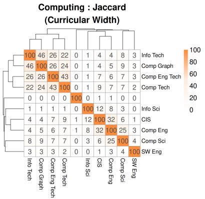




- Animal Sci (CIP 01) clusters with Biochemistry & Animal Bio (CIP 26)
- Strongest PMI: Animal Sci–Biochem (~23%)
- Crosses CIP family boundaries
- Math–Chem TF-IDF (~49%): strongest specialized pair

Computing: The Nested Hierarchy





- IT–CS: cleanest nested example in the dataset
  - Overlap: ~80%
  - Jaccard: ~8.5%
  - Largest divergence between these metrics anywhere
- CS–CompE: mixed nested/convergent (Overlap ~52%, TF-IDF ~29%)
- Computing is organized as *nested hierarchies* branching from CS

Interactive Explorer

- All metrics, domains, and program pairs
- Heatmaps with dendrograms
- Computing, engineering, sciences, cross-domain

https://ransomts.github.io/ASEE-2026/

Top Nested Relationships

High Overlap + Low Jaccard: one program's coursework is largely contained within another

Pair	Domain	Jaccard	TF-IDF	Overlap
Aerospace Eng. – Environmental Eng.	Eng.	8.2%	24.6%	90.3%
Textile Sci. – Environmental Eng.	Eng.	6.3%	2.6%	85.1%
Construction Tech. – Environmental Eng.	Eng.	6.8%	2.8%	80.9%

- Environmental Engineering's small course footprint is almost entirely shared with larger programs
- Low Jaccard because the larger programs have many additional courses

Top Parallel Relationships

Medium Jaccard + Low TF-IDF: shared foundational base, divergent specialization

Pair	Domain	Jaccard	TF-IDF	Overlap
Petroleum Eng. – Engineering Physics	Eng.	34.7%	2.7%	58.7%
Mechanical Eng. – Math	Cross-domain	35.9%	6.2%	57.3%
Computer Tech. – Engineering Physics	Eng.	32.6%	3.0%	49.3%

- Shared prerequisites and gen-ed courses drive moderate Jaccard
- Near-zero TF-IDF means their *identity-defining* coursework is completely different

Top Convergent Relationships

High Jaccard + High TF-IDF + High Overlap: deep structural alignment

Pair	Domain	Jaccard	TF-IDF	Overlap
Computer Eng. – Construction Eng. Tech.	Eng.	47.1%	55.3%	78.8%
Environmental Design – BA Env. Studies	Sciences	47.3%	45.8%	70.8%
Chemistry – BA Env. Studies	Sciences	46.1%	45.6%	72.0%

- These pairs share both the same courses *and* similar enrollment frequencies
- Often not grouped administratively despite deep curricular overlap

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Key Takeaways

- STEM domains are interdisciplinary in *structurally different ways*
  - Engineering: parallel (shared core, divergent specialization)
  - Computing: nested hierarchies from CS
  - Sciences: cross-boundary clusters
- One-size-fits-all policy won't work

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Practical Implications

- Nested (IT $\subset$ CS): guide students toward parent field's electives
- Parallel (engineering): coordinate shared prerequisites
- Convergent (Aero $\approx$ Ind Eng): cross-listed courses, certificates, dual degrees

Future Work

- Disaggregate required vs. elective coursework
- Time series: are programs converging or diverging?
- Institutional disaggregation across universities
- Validate against known interdisciplinary programs

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